

5 Entire Functions

...but after the 15th of October I felt myself a free man, with such longing for mathematical work, that the last two months flew by quickly, and that only today I found the letter of the 19th of October that I had not answered. The result of my work, with which I am not entirely satisfied, I want to share with you.

Firstly, in looking back at my lectures, a gap in function theory needed to be filled. As you know, up to now the following question had been unresolved. Given an arbitrary sequence of complex numbers, $a_1, a_2, \dots$, can one construct an entire (transcendental) function that vanishes at these values, with prescribed multiplicities, and nowhere else?...

K. Weierstrass, 1874

In this chapter, we will study functions that are holomorphic in the whole complex plane; these are called **entire** functions. Our presentation will be organized around the following three questions:

1. Where can such functions vanish? We shall see that the obvious necessary condition is also sufficient: if $\{z_n\}$ is any sequence of complex numbers having no limit point in $\mathbb{C}$, then there exists an entire function vanishing exactly at the points of this sequence. The construction of the desired function is inspired by Euler's product formula for $\sin \pi z$ (the prototypical case when $\{z_n\}$ is $\mathbb{Z}$), but requires an additional refinement: the Weierstrass canonical factors.
2. How do these functions grow at infinity? Here, matters are controlled by an important principle: the larger a function is, the more zeros it can have. This principle already manifests itself in the simple case of polynomials. By the fundamental theorem of algebra, the number of zeros of a polynomial P of degree d is precisely d , which is also the exponent in the order of (polynomial) growth of P , namely

$$\sup_{|z|=R} |P(z)| \approx R^d \quad \text{as } R \rightarrow \infty.$$

A precise version of this general principle is contained in Jensen's formula, which we prove in the first section. This formula, central to much of the theory developed in this chapter, exhibits a deep connection between the number of zeros of a function in a disc and the (logarithmic) average of the function over the circle. In fact, Jensen's formula not only constitutes a natural starting point for us, but also leads to the fruitful theory of value distributions, also called Nevanlinna theory (which, however, we do not take up here).

3. To what extent are these functions determined by their zeros? It turns out that if an entire function has a finite (exponential) order of growth, then it can be specified by its zeros up to multiplication by a simple factor. The precise version of this assertion is the Hadamard factorization theorem. It may be viewed as another instance of the general rule that was formulated in Chapter 3, that is, that under appropriate conditions, a holomorphic function is essentially determined by its zeros.

1 Jensen's formula

In this section, we denote by D_R and C_R the open disc and circle of radius R centered at the origin. We shall also, in the rest of this chapter, exclude the trivial case of the function that vanishes identically.

Theorem 1.1 *Let Ω be an open set that contains the closure of a disc D_R and suppose that f is holomorphic in Ω , $f(0) \neq 0$, and f vanishes nowhere on the circle C_R . If $z_1, \dots, z_N$ denote the zeros of f inside the disc (counted with multiplicities),¹ then*

$$(1) \quad \log |f(0)| = \sum_{k=1}^N \log \left(\frac{|z_k|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta.$$

The proof of the theorem consists of several steps.

Step 1. First, we observe that if f_1 and f_2 are two functions satisfying the hypotheses and the conclusion of the theorem, then the product $f_1 f_2$ also satisfies the hypothesis of the theorem and formula (1). This observation is a simple consequence of the fact that $\log xy = \log x + \log y$ whenever x and y are positive numbers, and that the set of zeros of $f_1 f_2$ is the union of the sets of zeros of f_1 and f_2 .

¹That is, each zero appears in the sequence as many times as its order.

Step 2. The function

$$g(z) = \frac{f(z)}{(z - z_1) \cdots (z - z_N)}$$

initially defined on $\Omega - \{z_1, \dots, z_N\}$, is bounded near each z_j . Therefore each z_j is a removable singularity, and hence we can write

$$f(z) = (z - z_1) \cdots (z - z_N)g(z)$$

where g is holomorphic in Ω and nowhere vanishing in the closure of D_R . By Step 1, it suffices to prove Jensen's formula for functions like g that vanish nowhere, and for functions of the form $z - z_j$.

Step 3. We first prove (1) for a function g that vanishes nowhere in the closure of D_R . More precisely, we must establish the following identity:

$$\log |g(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |g(Re^{i\theta})| d\theta.$$

In a slightly larger disc, we can write $g(z) = e^{h(z)}$ where h is holomorphic in that disc. This is possible since discs are simply connected, and we can define $h = \log g$ (see Theorem 6.2 in Chapter 3). Now we observe that

$$|g(z)| = |e^{h(z)}| = |e^{\operatorname{Re}(h(z)) + i \operatorname{Im}(h(z))}| = e^{\operatorname{Re}(h(z))},$$

so that $\log |g(z)| = \operatorname{Re}(h(z))$. The mean value property (Corollary 7.3 in Chapter 3) then immediately implies the desired formula for g .

Step 4. The last step is to prove the formula for functions of the form $f(z) = z - w$, where $w \in D_R$. That is, we must show that

$$\log |w| = \log \left(\frac{|w|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |Re^{i\theta} - w| d\theta.$$

Since $\log(|w|/R) = \log |w| - \log R$ and $\log |Re^{i\theta} - w| = \log R + \log |e^{i\theta} - w/R|$, it suffices to prove that

$$\int_0^{2\pi} \log |e^{i\theta} - a| d\theta = 0, \quad \text{whenever } |a| < 1.$$

This in turn is equivalent (after the change of variables $\theta \mapsto -\theta$) to

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = 0, \quad \text{whenever } |a| < 1.$$

To prove this, we use the function $F(z) = 1 - az$, which vanishes nowhere in the closure of the unit disc. As a consequence, there exists a holomorphic function G in a disc of radius greater than 1 such that $F(z) = e^{G(z)}$. Then $|F| = e^{\operatorname{Re}(G)}$, and therefore $\log |F| = \operatorname{Re}(G)$. Since $F(0) = 1$ we have $\log |F(0)| = 0$, and an application of the mean value property (Corollary 7.3 in Chapter 3) to the harmonic function $\log |F(z)|$ concludes the proof of the theorem.

From Jensen's formula we can derive an identity linking the growth of a holomorphic function with its number of zeros inside a disc. If f is a holomorphic function on the closure of a disc D_R , we denote by $\mathbf{n}(r)$ (or $\mathbf{n}_f(r)$ when it is necessary to keep track of the function in question) the number of zeros of f (counted with their multiplicities) inside the disc D_r , with $0 < r < R$. A simple but useful observation is that $\mathbf{n}(r)$ is a non-decreasing function of r .

We claim that if $f(0) \neq 0$, and f does not vanish on the circle C_R , then

$$(2) \quad \int_0^R \mathbf{n}(r) \frac{dr}{r} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

This formula is immediate from Jensen's equality and the next lemma.

Lemma 1.2 *If $z_1, \dots, z_N$ are the zeros of f inside the disc D_R , then*

$$\int_0^R \mathbf{n}(r) \frac{dr}{r} = \sum_{k=1}^N \log \left| \frac{R}{z_k} \right|.$$

Proof. First we have

$$\sum_{k=1}^N \log \left| \frac{R}{z_k} \right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}.$$

If we define the characteristic function

$$\eta_k(r) = \begin{cases} 1 & \text{if } r > |z_k|, \\ 0 & \text{if } r \leq |z_k|, \end{cases}$$

then $\sum_{k=1}^N \eta_k(r) = \mathbf{n}(r)$, and the lemma is proved using

$$\sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r} = \sum_{k=1}^N \int_0^R \eta_k(r) \frac{dr}{r} = \int_0^R \left(\sum_{k=1}^N \eta_k(r) \right) \frac{dr}{r} = \int_0^R \mathbf{n}(r) \frac{dr}{r}.$$

2 Functions of finite order

Let f be an entire function. If there exist a positive number ρ and constants $A, B > 0$ such that

$$|f(z)| \leq Ae^{B|z|^\rho} \quad \text{for all } z \in \mathbb{C},$$

then we say that f has **an order of growth** $\leq \rho$. We define *the order of growth of f* as

$$\rho_f = \inf \rho,$$

where the infimum is over all $\rho > 0$ such that f has an order of growth $\leq \rho$.

For example, the order of growth of the function e^{z^2} is 2.

Theorem 2.1 *If f is an entire function that has an order of growth $\leq \rho$, then:*

- (i) $n(r) \leq Cr^\rho$ for some $C > 0$ and all sufficiently large r .
- (ii) If $z_1, z_2, \dots$ denote the zeros of f , with $z_k \neq 0$, then for all $s > \rho$ we have

$$\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty.$$

Proof. It suffices to prove the estimate for $n(r)$ when $f(0) \neq 0$. Indeed, consider the function $F(z) = f(z)/z^\ell$ where ℓ is the order of the zero of f at the origin. Then $n_f(r)$ and $n_F(r)$ differ only by a constant, and F also has an order of growth $\leq \rho$.

If $f(0) \neq 0$ we may use formula (2), namely

$$\int_0^R n(x) \frac{dx}{x} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

Choosing $R = 2r$, this formula implies

$$\int_r^{2r} n(x) \frac{dx}{x} \leq \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

On the one hand, since $n(r)$ is increasing, we have

$$\int_r^{2r} n(x) \frac{dx}{x} \geq n(r) \int_r^{2r} \frac{dx}{x} = n(r) [\log 2r - \log r] = n(r) \log 2,$$